

Section A

- 1 (a) Fig 1.1 shows the first ionisation energies of calcium to rubidium.

Use the *Data Booklet* and your knowledge of the variation in first ionisation energy of the elements of Period 3 (Na to Ar) to estimate and plot, on Fig 1.1, the first ionisation energies of the elements: Ga, As and Kr.

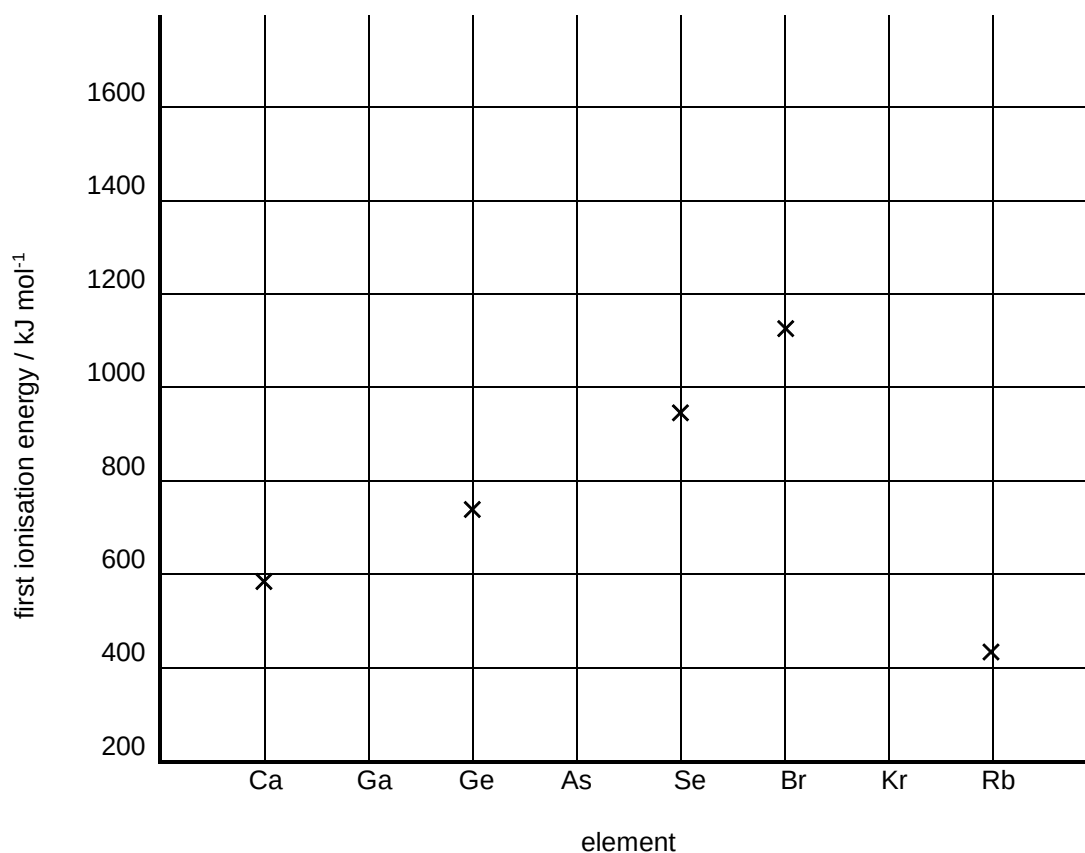
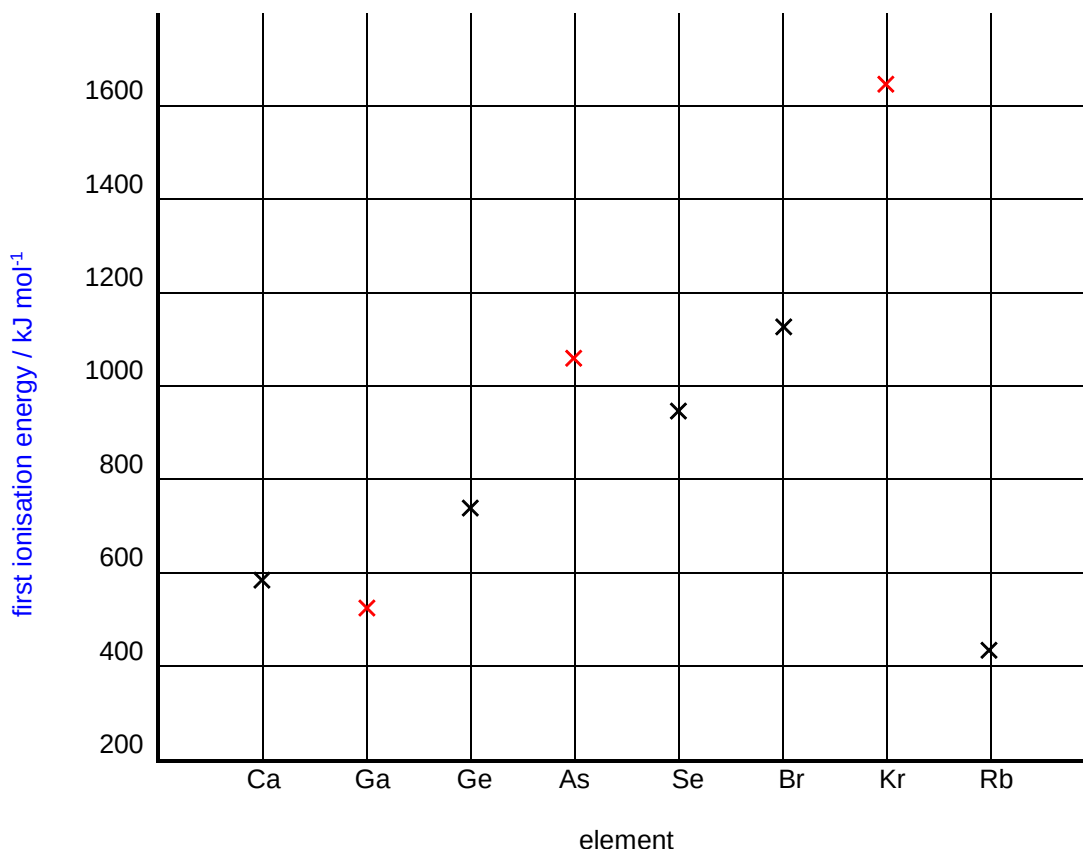


Fig 1.1

[2]



Rb < 1st IE of Ga < Ca (value can be obtained from Data Booklet)

Se < 1st IE of As < Br

Br < 1st IE of Kr

- (b) (i) Justify your prediction in (a), by explaining why the first ionisation energy of As you have plotted is higher/lower/similar to Ge and Se. [3]

First IE of As is higher than Ge. As has higher nuclear charge but similar shielding effect as Ge, hence, the electrostatic attraction between its valence electron is stronger than Ge.

First IE of Se is lower than As. The first electron (valence electron) to remove from Se is a paired electron with inter-electronic repulsion. Hence, the electrostatic attraction between its valence electron is weaker than As and required less energy to remove.

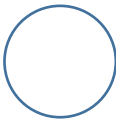
- (ii) Explain why the first ionisation energy of Kr you have plotted is higher/lower/similar to Rb. [1]

First IE of Kr is higher than Rb.

The 5s electron to be removed from Rb is at a higher energy level/ further away from the nucleus than the 4p electron in Kr. This 5s electron experiences more shielding due to more inner electrons despite increased in nuclear charge.

The 5s electron in Rb is less strongly attracted by the nucleus and requires less energy to be removed compared to the 4p electron in Kr.

- (c) Draw the shape of the highest occupied orbital in Ca and Ga atom. [2]

Ca	Ga
 4s (no need to label)	 4p

- (d) A student conducted some experiments involving Period 3 oxides and recorded his observations.

element	observations
W	Oxide of W is a gas which dissolves completely in water to form a solution which turns blue litmus paper red.
X	Oxide of X is a white solid which dissolves completely in water to form a dark blue solution in presence of Universal Indicator.
Y	Oxide of Y does not dissolve in water and but reacts with concentrated NaOH on heating.
Z	Oxide of Z can react with both aq NaOH and aq HCl.

- (i) Identify the unknown elements. [2]

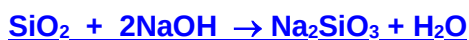
Unknown element	Identity
W	<u>Sulfur / S</u>
X	<u>Sodium / Na</u>
Y	<u>Silicon / Si</u>
Z	<u>Aluminium / Al</u>

- (ii) Oxide of **W** and oxide of **X** are mixed together in the same beaker of water. State the type of reaction that will occur. [1]

Acid-base / neutralisation reaction.

Reason (not required in students' answer): oxide of non-metal, W (SO₂) is acidic and oxide of metal, X (Na₂O) is basic; thus it will be acid-base reaction.

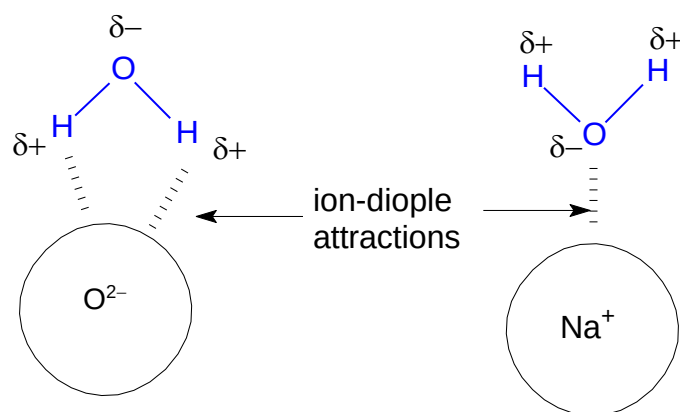
- (iii) Use the data given to write a chemical equation to show how the oxides of **Y** reacts with sodium hydroxide. [1]



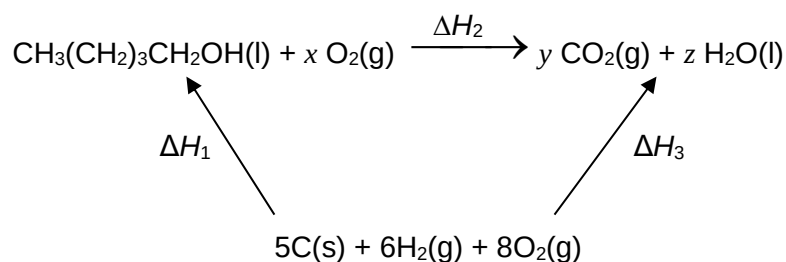
OR



- (iv) Draw 2 simple diagrams to show how a water molecule can be attached to a sodium cation, and to an oxide anion. Label each diagram to show the type of interaction involved. [2]



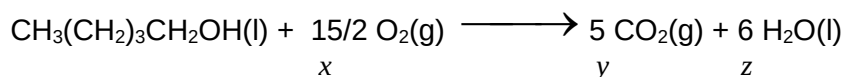
- 2 (a) Consider the following energy cycle involving pentan-1-ol, $\text{CH}_3(\text{CH}_2)_3\text{CH}_2\text{OH}$.



- (i) Define the term *standard enthalpy change of combustion* of pentan-1-ol. [1]

(Heat) Energy evolved when 1 mole of pentan-1-ol is completely burnt in excess oxygen at standard conditions of 1 bar and 298K.

- (ii) State the values of x , y and z . [1]



- (iii) What does ΔH_1 represent? [1]

ΔH_1 represent enthalpy change of formation of 1 mol pentan-1-ol, $\text{CH}_3(\text{CH}_2)_3\text{CH}_2(\text{OH})$ from its elements in their standard states.

- (iv) Using the following data, calculate ΔH_1 .

enthalpy change of combustion of carbon	= -393 kJ mol ⁻¹	
enthalpy change of combustion of hydrogen	= -285 kJ mol ⁻¹	
enthalpy change of combustion of pentan-1-ol	= -3331 kJ mol ⁻¹	[2]
$\Delta H_1 = \Delta H_3 - \Delta H_2$		
= 5(-393) + 6(-285) - (-3331)		
= <u>-344 kJ mol⁻¹</u>		

- (b) Ammonia, NH_3 is commonly used as a household cleaner to remove greasy dirt in ovens. It is also used to clean windows as it evaporates quickly and does not leave stains on glass.

25.0 cm^3 of 0.500 mol dm^{-3} NH_3 was titrated against dilute HNO_3 .

Fig. 2.1 shows the pH changes during the titration.

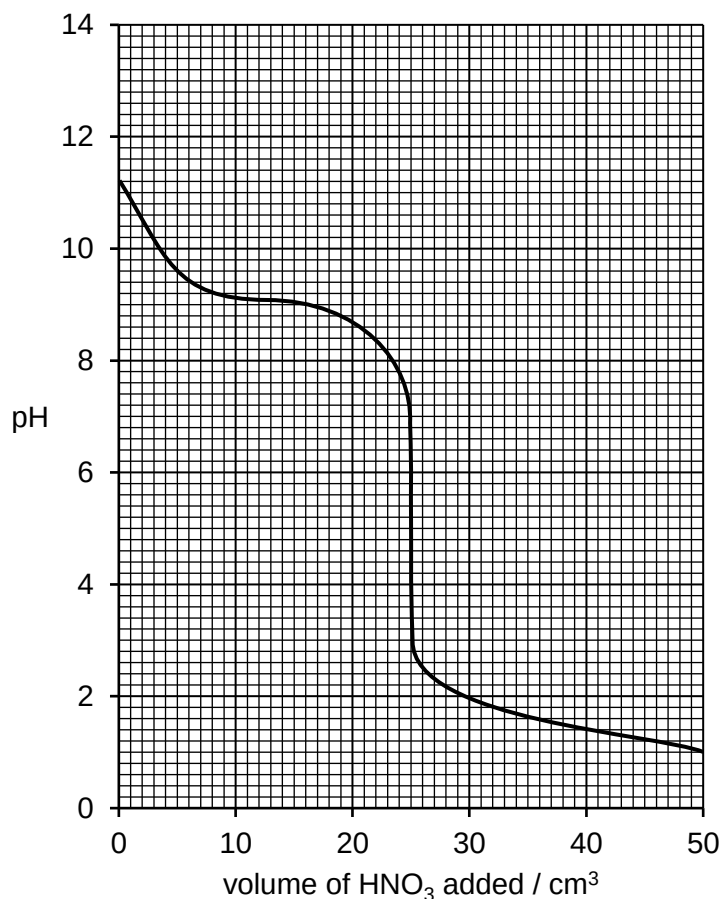


Fig. 2.1

- (i) Using the pH from Fig 2.1, calculate the concentration of hydroxide ions, in mol dm^{-3} , in 25.0 cm^3 of 0.500 mol dm^{-3} solution of NH_3 . [1]

$$\text{pOH} = 14 - 11.2 = 2.8$$

$$[\text{OH}^-] = 10^{-2.8} = 0.0015849 \text{ mol dm}^{-3}$$

$$\approx \underline{\underline{0.00158 \text{ mol dm}^{-3}}}$$

- (ii) Explain how your answer in (b)(i) indicates that NH_3 is a weak base. [1]

$$[\text{OH}^-] \ll [\text{NH}_3]$$

OR

The concentration of OH^- produced from the dissociation of NH_3 is lower than the initial concentration of NH_3 .

Hence only partial dissociation has occurred.

- (iii) Using Fig. 2.1, state the volume of nitric acid required to reach the equivalence point for this titration.

Hence, suggest the concentration of the nitric acid use in the titration. [1]

Volume of nitric acid required is 25 cm³.

Since mole ratio of NH₃ and HNO₃ is 1: 1, [HNO₃] = 0.500 mol dm⁻³

- (iv) Table 2.1 shows some acid-base indicators.

Table 2.1

indicator	colour in acid	colour in alkali	pH range of colour change of indicator
alizarin yellow	yellow	orange	10.1 – 13.0
methyl orange	red	yellow	3.1 – 4.4
phenolphthalein	colourless	pink	8.2 – 10.0

Suggest which of the above indicator should be used for this titration.

State the colour change of your chosen indicator at the end-point of this titration.

indicator

colour change [2]

methyl orange

yellow to orange

- 3 (a) Vapour of 0.2 mol phosphorus(III) chloride, 0.15 mol chlorine and 0.3 mol phosphorus(V) chloride can exist in equilibrium in a 5.0 dm³ vessel at 230 °C.



- (i) Calculate the value of K_c at 230 °C, stating its unit. [2]

$$K_c = \frac{(0.3/5)}{(0.2/5)(0.15/5)} = 50.0 \text{ mol}^{-1}\text{dm}^3$$

- (ii) Suggest, with reasoning, whether lowering the temperature would favour the formation of PCl_5 . [1]

When temperature is lowered, position of equilibrium shifts to the right to favour forward exothermic reaction to produce more heat, hence producing more PCl_5 .

- (b) Phosphorus(V) chloride dissolves in polar solvents forming two ions $[\text{PCl}_4]^+$ and $[\text{PCl}_6]^-$. Draw the dot-and-cross diagram and shape of $[\text{PCl}_4]^+$. [2]

Ion	Dot-and-cross diagram	Diagram of shape
$[\text{PCl}_4]^+$		

- 4 (a) Carbon particles, oxides of carbon and nitrogen are commonly produced in the internal combustion engines of cars. These substances have significant environmental consequences if emitted into the atmosphere. Hence, catalytic converter is fitted to the exhaust to reduce the amount of polluting substances emitted.

Fig 4.1 shows a catalytic converter that include a particle filter to remove solid particles of carbon.

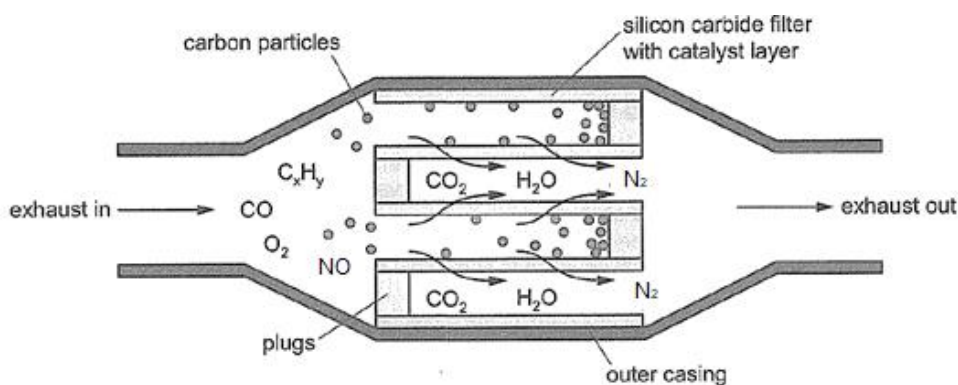
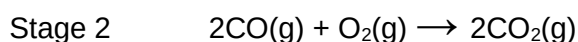
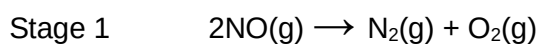


Fig 4.1

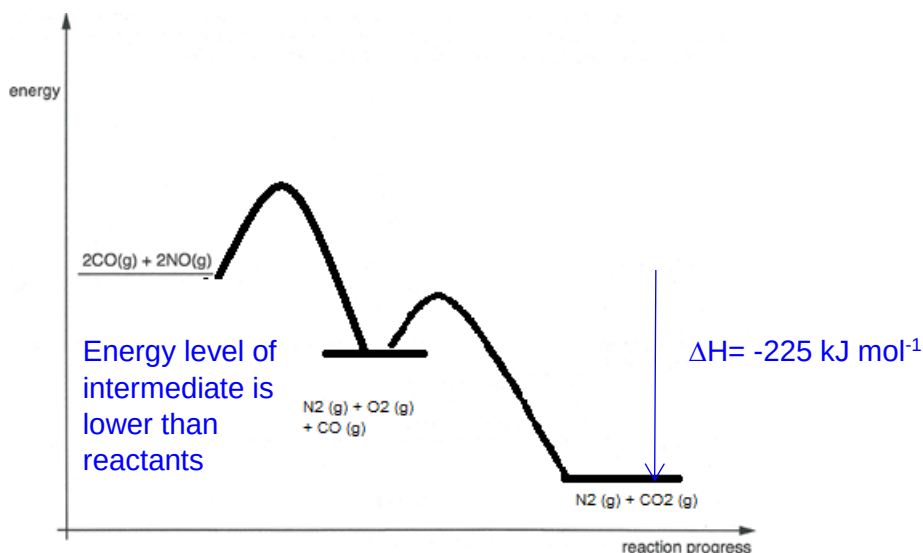
- (i) The catalytic converter can remove nitrogen monoxide from the exhaust gases along with carbon monoxide, CO, which is also formed.

The conversion of NO and CO into N_2 and CO_2 , which occurs in two stages has an overall $\Delta H_{\text{reaction}} = -225 \text{ kJ mol}^{-1}$. The oxygen produced in stage 1 is used in stage 2. Both stages are exothermic

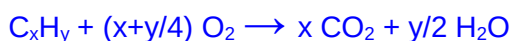


Complete the reaction pathway diagram on Fig 4.2 for the overall reaction between NO and CO which involves both of these stages.

[2]



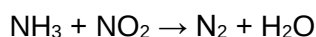
- (ii) Write an equation to show how unburnt hydrocarbon, C_xH_y can be removed. [1]



- (iii) Suggest why nitrogen monoxide is produced when petrol is burned in the internal combustion engine but not produced when petrol is burned in a spirit burner in a laboratory. [1]

Temperature in the car engine is high enough to break the strong triple bond in N_2 for combustion of N_2 to occur.

- (b) Diesel engines may also be fitted with a second catalytic converter to remove the remaining oxides of nitrogen. The second converter uses ammonia as the reducing agent. One possible reaction is shown by the following **unbalanced** equation.

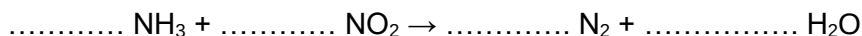


- (i) Deduce all the changes in oxidation number, stating which element is involved in the change. [1]

Oxidation number of N in NH_3 changes from -3 to 0 in N_2 .

Oxidation number of N in NO_2 changes from +4 to 0 in N_2 .

- (ii) Hence, balance the equation.



[1]

- (c) Silicon carbide, SiC, is strong and resistant to melting at high temperature of the exhaust. Hence, it is a suitable material use for making the filter and for coating the Pt catalyst.

- (i) Suggest how the structure and bonding in SiC gives its properties above. [2]

Giant covalent structure

Extensive network of strong covalent bonding between Si and C atoms
needs a lot of energy to overcome.

This gives its strength and high melting point.

- CLT (ii) Clusters of platinum catalyst coated onto fine powder of SiC in the converter is found to be around 4 nm in size.

Explain how this greatly increase the catalytic activity of the Pt catalyst. [2]

The Pt is a nanoparticle since it is between 1-100 nm.

With this dimension, the Pt particles have greater/larger surface area to volume ratio, resulting in a higher proportion of the particles at the surface being able to react leading to increased antibacterial activity.

- 5 (a) Infra-red spectroscopy is a fast, reliable and cost-effective technique suitable for analysis of polymers. The infra-red absorption frequencies in the spectrum give information about the presence or absence of specific functional groups in a molecule and can be used to determine the identity of the sample.

Three samples of polymers namely poly(vinyl alcohol) (PVA), polyethylene terephthalate (PET) and polyacrylamide (PAM) were taken and analysed using infra-red spectroscopy.

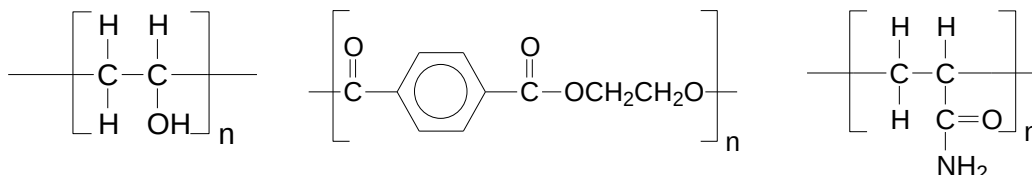


Table 5.1 shows some infra-red absorption frequencies in the spectra obtain from the three polymers.

Table 5.1

Spectrum	Infra-red absorption frequencies/cm ⁻¹
A	3100, 1740, 1050
B	3300, 1690
C	3200, 1100

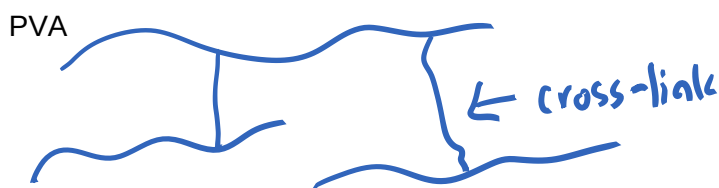
Using relevant data from *Data Booklet*, assign the identity of the three polymers to the relevant spectrum. Explain your answer. [6]

Spectrum	Polymer	Reasoning
A	PET	3100 suggests presence of benzene Or 1740 suggests presence of C=O in ester. Or 1740 <u>and</u> 1050 suggest presence of C=O and C-O in ester. [Do not accept only 1050 use for C-O bond as they could be due to C-O in PVA.]
B	PAM	1690 suggests presence of C=O in amide. Or 3300 suggests presence of N-H.
C	PVA	3200 suggests presence of O-H. Or 1100 suggests presence of C-O in alcohol.

- (b) PVA can be cross-linked with pentane-1,5-dial to form a three-dimensional network of polymer known as hydrogel. The hydrogel can swell in water and hold a large amount of water while maintaining the structure. Water is a large constituents of human body. Hence, hydrogel can be used for drug delivery in pharmaceutical field.

- (i) Draw a simple well-labelled diagram to present the cross-links in hydrogel made from PVA and pentane-1,5-dial.

[Structural formulae of molecules are **not necessary** in your diagram.] [2]



- (ii) Explain, in terms of the structure and bonding, the properties exhibited by hydrogel? [2]

Presence of -OH groups in PVA able to form hydrogen bondings with water molecules hence making it hydrophilic.

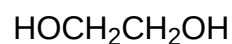
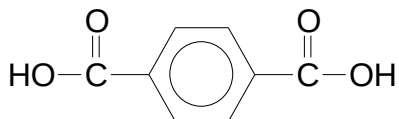
Cross-linked covalent bonds strengthen the structure of the polymer giving the 3D structure. Though the polymer can swell, the structure maintains due to its strength.

- (iii) Suggest whether hydrogel is classified as thermoplastic or thermosetting polymer. [1]

Thermosetting as cross-links prevent the polymers from melting.
At high temperature, it decomposes.

- (c) (i) State the type of polymerisation used to form PET. Hence, suggest the structures of the two monomers use to form PET.

Condensation polymerisation



- (ii) PET is commonly used as single-use food packaging. Explain why PET are unsuitable for use in bottles storing pickles in acidic vinegar in terms of its structure and bonding. [1]

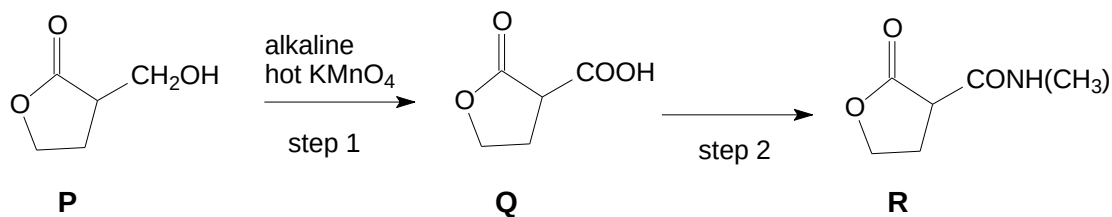
PET contains ester functional groups in the polymer, which can be hydrolysed by the acidic vinegar.

- (d) State one negative impact from the use of non-recyclable plastics. [1]

Plastics are finite resources (economic factor)
Non-biodegradable resulting in landfill capacity exhausted
Or any sound impact.

6

A student proposed the following reaction scheme to synthesise **R** from **P**.



(a) State the types of reaction in steps 1 and 2. [2]

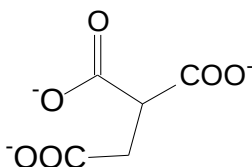
step 1 : Oxidation
 step 2 : condensation

(b) Suggest the reagents and conditions for step 2. [1]

CH₃NH₂ with DCC

(c) The student found that compound **Q** was not obtained as a major product in step 1 in the reaction scheme. Instead, compound **S** was obtained.

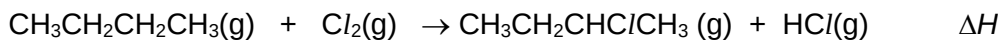
Suggest the structure of compound **S**. [2]



Section B

Answer **one** question from this section, in the spaces provided.

- 7 (a) When briefly exposed to a bright light, a mixture of butane gas and chlorine gas reacts to give 2-chlorobutane and hydrogen chloride gas.

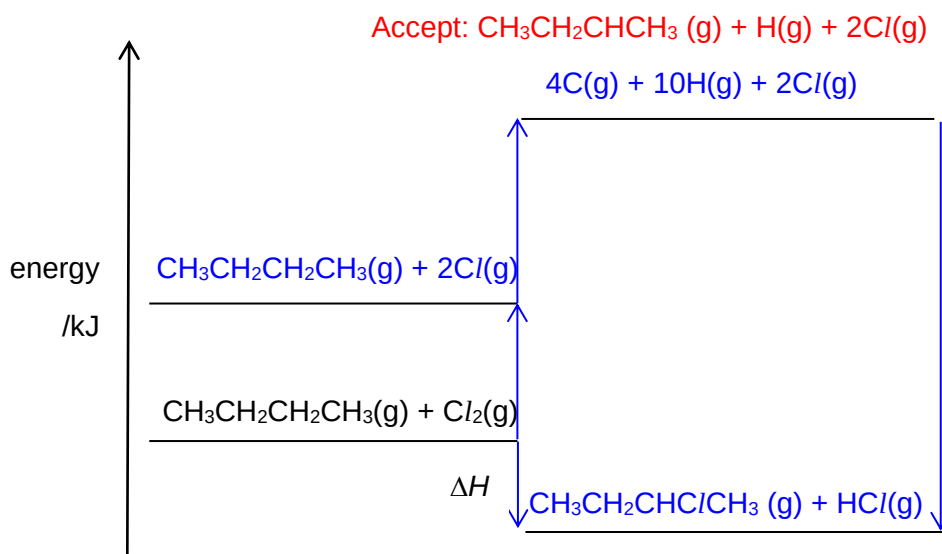


- (i) Use relevant bond energy values given in the *Data Booklet* to calculate the enthalpy change for this reaction, ΔH . [1]

$$\begin{aligned} \Delta H &= \text{BE (Bonds broken)} - \text{BE (Bonds formed)} \\ &= [\text{BE (C-H)} + \text{BE (Cl-Cl)}] - [\text{BE (H-Cl)} + \text{BE (C-Cl)}] \\ &= [410 + 244] - [431 + 340] \\ &= \underline{\underline{-117 \text{ kJ mol}^{-1}}} \end{aligned}$$

- (ii) Draw arrows on the energy diagram below to show each of the bond breaking and bond making steps you have used in your calculation above.

Label each level with the appropriate formulae. [3]



- (iii) Suggest why the reaction only starts when it is exposed to bright light. [1]
 Light is required to provide energy to overcome the activation energy so as to kick start the reaction.

OR

Light is required to provide energy to break the Cl-Cl bond, forming the Cl radicals in order to start the reaction.

- (iv) State the type of reaction between butane and chlorine. [1]

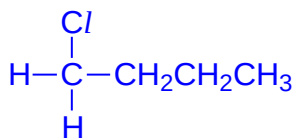
(Free radical) substitution reaction

- (v) Suggest why the reaction continues with **increasing** rate after the **brief** exposure to the bright light has stopped. [1]

Since the reaction is exothermic, the heat produced causes an increase in temperature [1]. Thus, the rate of reaction increases.

- (vi) In the reaction of butane with chlorine, another mono-chlorinated compound **S** is formed. Draw the structural formulae of this mono-chlorinated compound **S**.

Hence, suggest the statistical ratio of this compound **S** and 2-chlorobutane? [2]



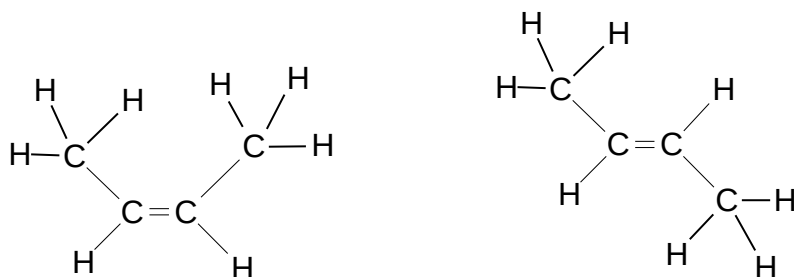
Compound **S**

ratio of Compound **S** : 2-chlorobutane = 3 : 2

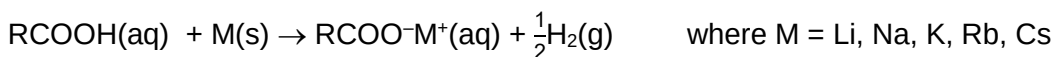
- (b) Under suitable conditions, 2-chlorobutane can be converted into two structural isomers of butene, of which one exhibit cis-trans isomerism.

- Suggest the reagent and condition for this reaction.
- Draw the displayed formula of the pair of cis-trans isomers. [3]

reagent and conditions: ethanolic NaOH/ KOH, Heat



- (c) Carboxylic acid reacts with Group 1 metals in the following reaction.



- (i) Suggest how you would expect the reactivity of Group 1 metals with carboxylic acid to change down the group. Explain your answer in terms of its reducing ability. [3]

The reactivity of Group 1 metals with carboxylic acid will increase down the group.

Down the group,

- Atomic radius increases / valence electrons are further away from the nucleus/ number of filled electron shells increases.
- There is weaker electrostatic forces of attraction between the nucleus and valence electrons
- Metal loses their valence electrons to form cation more easily

Metal have greater tendency to be oxidised

- (ii) Suggest with reasoning, whether you expect the same volume of hydrogen gas to be evolved in each case when excess carboxylic acid of same concentration is added separately to 5 g of different group 1 metal powders.

[No calculation required.] [3]

Going down group 1 elements, the volume of hydrogen gas evolved will decrease.

The limiting reagent is Group 1 metal.

Going down Group 1 elements, the atomic mass increases hence the number of moles of Group 1 metal will decreased, resulting in less moles of hydrogen gas evolved.

- (iii) Group 1 metals are often used in borohydrides, LiBH_4 , NaBH_4 and KBH_4 , as reducing agents in organic chemistry.

The greater the charge density of the metal cation, the stronger the reducing power of the borohydrides.

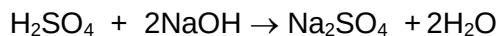
Identify the strongest reducing agent among the three Group 1 borohydrides listed above and explain your answer. [2]

Strongest reducing agent : LiBH_4

Among the Group 1 cations, all 3 ions have the same charge of +1 but Li^+ has the smallest ionic radius.

Li^+ has the largest charge density and hence, LiBH_4 is the strongest reducing agent.

- 8 (a) In an experiment to determine the maximum change in temperature when sulfuric acid is added to sodium hydroxide, 30 cm³ of sodium hydroxide is transferred to a polystyrene cup.



The initial temperature of the sodium hydroxide before sulfuric acid is added is measured and recorded.

5.0 cm³ portions of aqueous 0.7 mol dm⁻³ sulfuric acid are added to the polystyrene cup. The temperature of the solution in the cup is measured after each addition. The maximum temperature occurs when complete neutralisation has been achieved.

The results are shown in Fig. 8.1

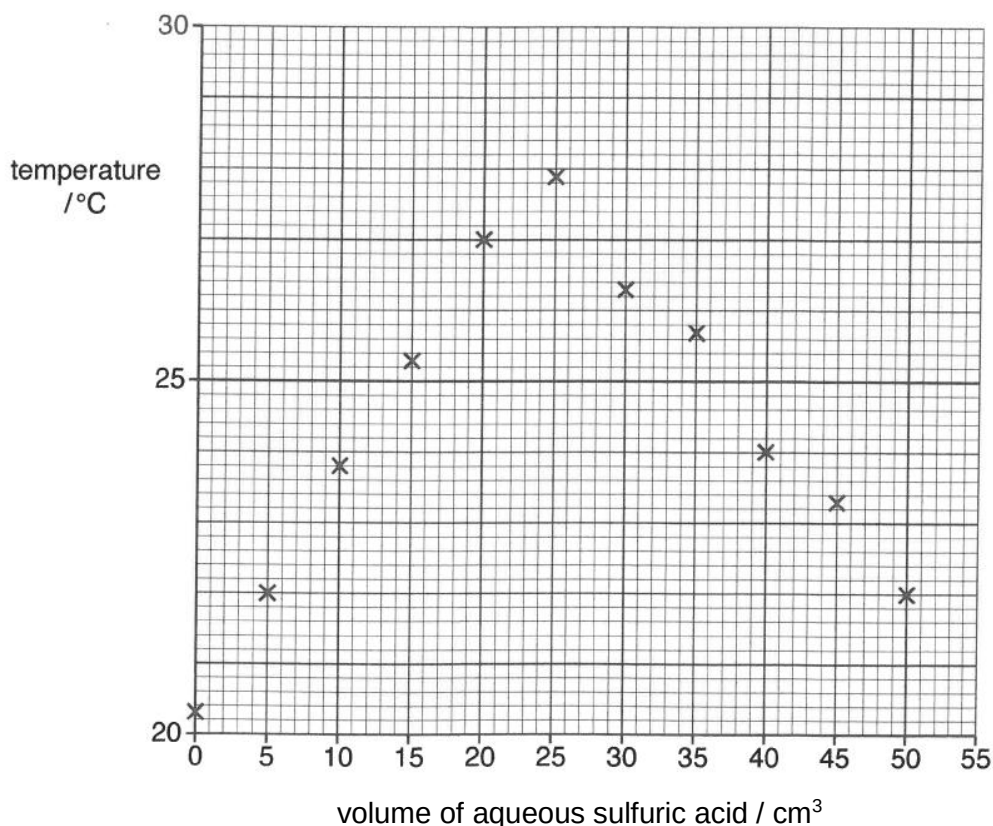
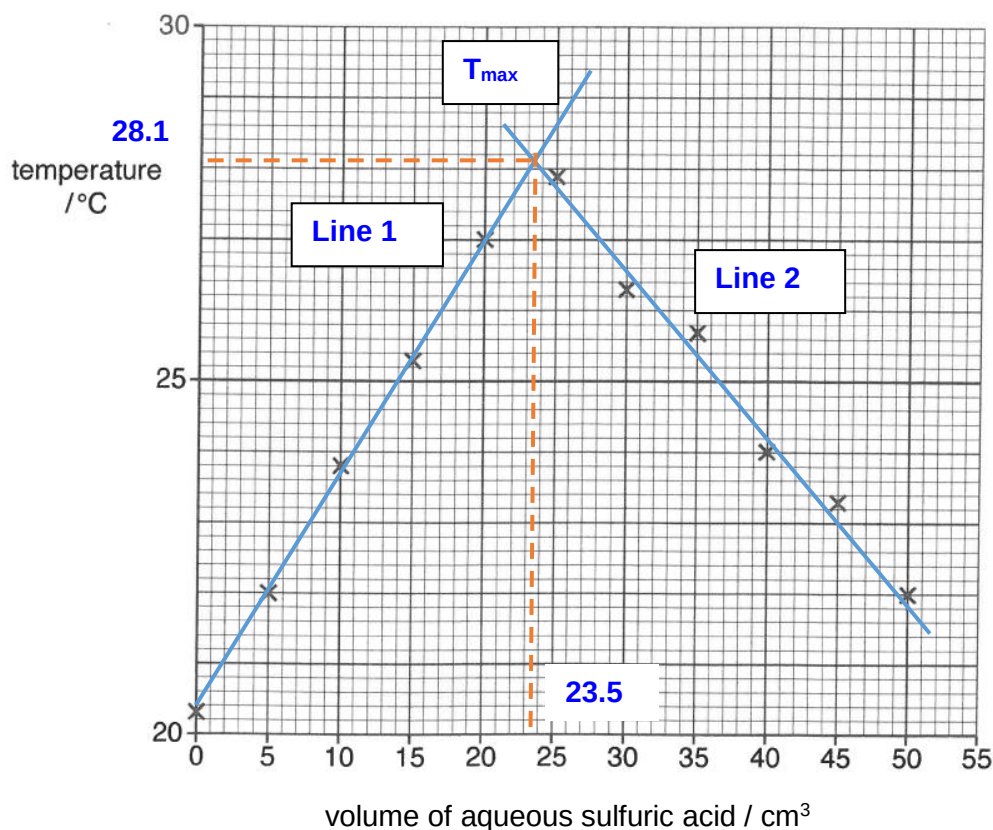


Fig. 8.1

- (i) Draw two separate straight lines of best fit on Fig 8.1.
 The first line should take into account the increasing temperatures.
 The second line should take into account the decreasing temperatures.
 Extend these lines until they intersect.
 Use your graph to determine the maximum temperature rise, ΔT_{max} , of the experiment. Show your working. [2]



Note: Random error is considered when drawing Line 2

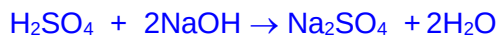
$T_{\max}$ is higher than all the points since it is an extrapolated value.

$$\Delta T_{\max} = 28.1 - 20.3 = \underline{7.8\text{ }^{\circ}\text{C}}$$

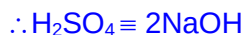
- (ii) Use your graph to determine the volume of sulfuric acid required to neutralise the sodium hydroxide and hence, determine the concentration of sodium hydroxide.

Show on your graph how you determined this volume. [3]

$$V_{\text{H}_2\text{SO}_4} = \underline{23.50\text{ cm}^3}$$



$$n_{\text{H}_2\text{SO}_4} = \frac{23.5}{1000} \times 0.7 = 0.01645\text{ mol}$$



$$n_{\text{NaOH}} = 2 \times 0.01645 = 0.0329\text{ mol}$$

$$[\text{NaOH}] = \frac{0.0329}{\frac{30}{1000}} = 1.097\text{ mol dm}^{-3} = 1.10\text{ mol dm}^{-3}$$

- (iii) Use your answers in (a)(i) and (a)(ii), calculate the energy, in kJ, evolved during the reaction.

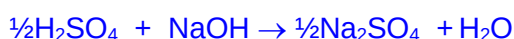
You should assume that the specific heat capacity of the reaction mixture is $4.18 \text{ J g}^{-1} \text{ K}^{-1}$ and the density of each solution is 1.00 g cm^{-3} . [1]

$$\begin{aligned}\text{Heat evolved} &= m C \Delta T_{\text{max}} \\ &= [(23.50 + 30.0) \times 1.00] \times 4.18 \times 7.8 \\ &= 1744 \text{ J} \\ &= \underline{\underline{1.74 \text{ kJ}}}\end{aligned}$$

- (iv) Using your answer to (a)(iii), calculate a value for the *enthalpy change of neutralisation*, in kJ mol^{-1} , for the reaction between sulfuric acid and sodium hydroxide. [1]

The point where Line 1 and Line 2 intersect shows the stoichiometry of the reaction between H_2SO_4 and NaOH .

Hence, number of moles of water formed = $n_{\text{NaOH}} = 0.0329 \text{ mol}$



$$\Delta H_{\text{neutralisation}} = -\frac{1.744}{0.0329} = -53.0 \text{ kJ mol}^{-1}$$

- (v) Suggest how the student's calculated value for *enthalpy change of neutralisation* in (a)(iv) differ from the literature value. [1]

The student's calculated value for enthalpy change of neutralisation in (a)(iv) will be less exothermic than the literature value.

- (b) When aqueous chlorine is added to an acidic solution containing Mn^{2+} ions, a black precipitate of MnO_2 is formed.

- (i) Write an equation for the reaction between Mn^{2+} ions and aqueous chlorine. [2]

Half-equations:



Overall Equations:



- (ii) Explain why this is a redox reaction, in terms of oxidation numbers. [1]

oxidation number of Cl : 0 to -1

oxidation number of Mn : +2 to +4

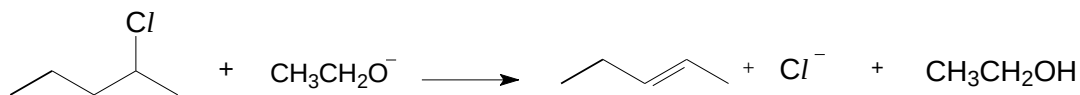
- (iii) There is no redox reaction when bromine is added to a solution containing Mn^{2+} ions. Explain the difference in observations between chlorine and bromine with Mn^{2+} in terms of ease of gaining of electrons. [3]

Chlorine is a stronger oxidising agent compared to bromine.

Chlorine has a smaller atomic radius than bromine, hence there are stronger electrostatic forces of attraction between the nucleus of chlorine and its valence electrons.

Therefore, ease of gaining electrons is higher for chlorine than bromine.

- (c) 2-bromopentane reacts with $\text{CH}_3\text{CH}_2\text{O}^-\text{Na}^+$ in the solvent ethanol according to the following equation.

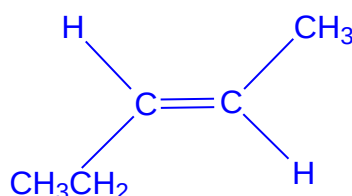
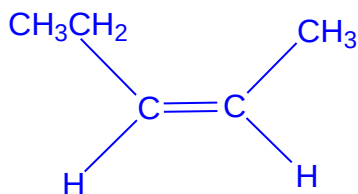


- (i) Name the type of reaction that occurred. [1]

Type of reaction: Elimination

- (ii) Draw the two isomers of the product pent-2-ene and name the type of isomerism they exhibit.

Explain why pent-2-ene show the type of isomerism, with reference to one of your diagrams. [4]



Type of isomerism : Cis-Trans Isomerism

There is restricted rotation about a C=C double bond and

two different groups attached to each C atom of the C=C double bond, CH_3CH_2 is different from H and CH_3 is different from H.

- (iii) State the observations when pent-2-ene reacts with aqueous bromine. [1]
Observations: Orange aqueous bromine decolourised.